Monotonic Sequences Worksheet

Name: _____ Date: _____

Type A: The Difference Test

Use the difference $a_{n+1} - a_n$ to show that the given sequence $\{a_n\}$ is strictly increasing or strictly decreasing.

---

1. $\left\{3 - \dfrac{2}{n}\right\}_{n=1}^{+\infty}$ ☐ strictly increasing ☐ strictly decreasing

---

2. $\left\{\dfrac{n}{5n-2}\right\}_{n=1}^{+\infty}$ ☐ strictly increasing ☐ strictly decreasing

---

3. $\{n - 3^n\}_{n=1}^{+\infty}$ ☐ strictly increasing ☐ strictly decreasing

Type B: The Ratio Test

Use the ratio a_{n+1}/a_n to show that the given sequence $\{a_n\}$ is strictly increasing or strictly decreasing. All terms of these sequences are positive.

---

4. $\left\{\dfrac{3^n}{2+3^n}\right\}_{n=1}^{+\infty}$ ☐ strictly increasing ☐ strictly decreasing

---

5. $\left\{\dfrac{n}{3^n}\right\}_{n=1}^{+\infty}$ ☐ strictly increasing ☐ strictly decreasing

---

6. $\left\{\dfrac{(2n)!}{(n!)^2}\right\}_{n=1}^{+\infty}$ ☐ strictly increasing ☐ strictly decreasing